

Diagnostic and monitoring applications using near infrared (NIR) spectroscopy in cancer and other diseases

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ABSTRACT

Early cancer diagnosis plays a critical role in improving treatment outcomes and increasing survival rates for certain cancers. NIR spectroscopy offers a rapid and cost-effective approach to evaluate the optical properties of tissues at the microvessel level and provides valuable molecular insights. The integration of NIR spectroscopy with advanced data-driven algorithms in portable instruments has made it a cutting-edge technology for medical applications. NIR spectroscopy is a simple, non-invasive and affordable analytical tool that complements expensive imaging modalities such as functional magnetic resonance imaging, positron emission tomography and computed tomography. By examining tissue absorption, scattering, and concentrations of oxygen, water, and lipids, NIR spectroscopy can reveal inherent differences between tumor and normal tissue, often revealing specific patterns that help stratify disease. In addition, the ability of NIR spectroscopy to assess tumor blood flow, oxygenation, and oxygen metabolism provides a key paradigm for its application in cancer diagnosis. This review evaluates the effectiveness of NIR spectroscopy in the detection and characterization of disease, particularly in cancer, with or without the incorporation of chemometrics and machine learning algorithms. The report highlights the potential of NIR spectroscopy technology to significantly improve discrimination between benign and malignant tumors and accurately predict treatment outcomes. In addition, as more medical applications are studied in large patient cohorts, consistent advances in clinical implementation can be expected, making NIR spectroscopy a valuable adjunct technology for cancer therapy management. Ultimately, the integration of NIR spectroscopy into cancer diagnostics promises to improve prognosis by providing critical new insights into cancer patterns and physiology.

1. Introduction

The continuous unregulated proliferation of cells can result in cancer, it is a diversified disease related to the cancer cell division, metastasis-associated profiles of tumors and organs related to the primary tumor. World Health Organization indicates cancer as the second most critical disease to cause death worldwide, it is related to -10 million deaths per year. The global number of cases is expected to be around 30 million cases in 2040, which is 47% higher from 2020 [1]. Efforts to

expand sustainable infrastructure for early cancer detection and evaluation of cancer therapy are imperative for global cancer control, especially in low- and middle-income countries [1].

In the context of sustainable technologies, the application of NIR spectroscopy for functional diagnosis and treatment monitoring of cancer has recently gained scientific attention. NIR spectroscopy is a simple, rapid, portable, non-invasive and affordable tool. Due to a known spectral window in the NIR range (650 to 950 nm), where tissue absorption is relatively low (up to several centimeters), the light can

Abbreviations: NIR, near infrared; FTIR, Fourier Transform Infrared; ATR, Attenuated Total Reflectance; PCA, Principal Component Analysis; PLS-DA, Partial Least Squares - Discriminant Analysis; PET, Positron emission tomography; CT, Computed tomography; fMRI, functional magnetic resonance imaging; PLSR, Partial Least Square Regression; FT-NIR, Fourier Transform Near infrared.

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penetrate thick or deep tissues [2].

NIR spectroscopy platforms are sensitive to detect changes in the shape of oxyhemoglobin or deoxyhemoglobin, lipid or water bands. NIR spectroscopy is also capable to detect molecular fingerprints with characteristic frequencies as diagnostic markers. Cancer detection using NIR spectroscopy is based on differences in NIR absorbing components between cancerous and normal tissue. In this context, this specific spectral profile can be a complementary tool for diagnosis [2]. Although NIR spectroscopy has a very limited spatial resolution for deep tissue analysis, it is a rapid and cost-effective method for determining the optical properties of tissues at the microvessel level. NIR spectroscopy can detect endogenous differences between tumor and normal tissues by examining tissue absorption, scattering, and concentrations of oxy-, deoxy-, and total hemoglobin, water, and lipids, as well as blood oxygen saturation. An early cancer diagnosis can significantly improve treatment outcomes and substantially increase survival rates for certain cancers. In this framework, ionizing or non-ionizing radiological techniques such as X-ray and positron emission tomography (PET), are currently used as diagnostic methods for cancer. Some of these imaging modalities, such as computed tomography (CT), PET, and functional magnetic resonance imaging (fMRI), have evolved in recent years, but their use is still limited [3,4].

Cancer treatment can alter the hemodynamic and metabolic state of the tumor, which affects the efficacy of subsequent therapies [5]. Because of their high metabolic demand and/or occasionally poor perfusion, tumors are evident for NIR spectroscopy in the presence of high hemoglobin and low oxygen saturation. The choice of anticancer drugs and optimal dosing depends on tissue oxygen and the measurement of tumor hemodynamic changes [6]. In this context, the changes in tumor hemodynamic and metabolic profile of the tumor can also be a supplementary tool for monitoring various cancer treatments [2]. Radiation and photodynamic therapy present limited effects in patients with hypoxic malignancies [3,4]. The functional assessment of tumor blood flow and metabolism should provide valuable data comparing previous, in the course of treatment and after cancer treatment, which can be useful for early prediction and evaluation of the cancer therapy effectiveness [6,7]. Although changes in tumor metabolic demand and blood perfusion of tumors can promote profitable data for cancer management, the lack of appropriate technology has limited routine measurement of tumor hemodynamics and metabolism during cancer treatment [6].

In this review, we sought to evaluate the effectiveness of NIR spectroscopy in the detection and characterization of diseases, especially in cancer applications coupled, or not, with chemometrics and machine-learning algorithms. In this context were selected studies related to the biomedical application of NIR with cancer and other diseases searched in MEDLINE/PubMed, Scopus, EMBASE, and Web-of-Science databases without language restriction or year of publication.

1.1. An introduction to NIR spectroscopy

Near Infrared (NIR) spectroscopy is a powerful analytical technique that uses the absorption of electromagnetic radiation in the near infrared region of the electromagnetic spectrum to provide valuable insight into the molecular composition of a sample. NIR spectroscopy is based on the principle of molecular vibrations, in particular the vibrations of chemical bonds in molecules [8]. The basic principles of molecular vibrations are based on the concept that chemical bonds behave like springs and have characteristic vibrational frequencies. These frequencies are determined by several factors, including the masses of the atoms involved in the bond, the strength of the bond itself, and the nature of the environment. As a result, different types of chemical bonds, such as those found in functional groups like carbonyls, hydroxyls and amines, have different vibrational frequencies [9].

Infrared absorption spectroscopy (IR), a technique closely related to NIR spectroscopy, takes advantage of the fact that molecules can absorb

infrared radiation at specific frequencies that correspond to the energy required to initiate vibrational transitions. The absorption of IR radiation by molecules follows selection rules based on changes in dipole moment during the vibrational transition. These selection rules determine which vibrational modes are allowed or forbidden in the absorption process, resulting in characteristic absorption peaks in the IR spectrum [10]. In the NIR region of the spectrum, from about 700 to 2500 nanometers, harmonics and combinations of fundamental modes become accessible for absorption. Harmonics occur when a molecule absorbs radiation twice the fundamental frequency. Similarly, combinations occur when two or more vibrational modes interact, resulting in absorption peaks at frequencies equal to their sum or difference. The occurrence of harmonics and combinations in the NIR region is due to the higher energy of NIR radiation compared to the fundamental frequencies of molecular vibrations. Therefore, the NIR region provides the opportunity to study these higher energy transitions, allowing detection and analysis of molecular vibrations that are not readily accessible in the mid-infrared region. By exploiting the absorption of NIR radiation by harmonics and combinations of molecular vibrations, NIR spectroscopy enables the identification and quantification of different functional groups and molecular species in a sample [8]. This analytical technique offers several advantages, including its non-destructive and non-invasive nature, rapid data acquisition, and the ability to analyze samples in different states, such as liquids, solids, and gasses. NIR platforms require the application of multivariate algorithms to describe the response of the NIR spectrum to the chemical or physical properties of samples used for calibration, predictive purposes, or as a classifier. NIR spectroscopy is based on the correlations between a measured spectral response and the harmonics of fundamental frequencies at infrared frequencies [11]. Depending on the amount of each analyte, absorbing molecules in the sample, and sample thickness, the harmonic oscillations occur at different frequencies measured in NIR spectroscopy. Therefore, the variations in the response of the near-infrared spectrometer are proportional to the changes in the concentrations of the chemical components or the physical properties (scattering/absorption properties) in each sample [11]. Fig. 1, as mentioned, represents a calculation of the absorbance spectrum based on the measurement of reflectance. Reflectance spectroscopy involves illuminating a sample with NIR radiation and measuring the intensity of the radiation reflected from the sample's surface.

1.2. Synthesized history of the study of infrared absorptions

Abney and Festing used photographic plates to measure the first near-infrared spectra in 1881. They produce the first spectra and accurately hypothesized that the absorptions were related to the chemical composition of the sample. William W. Coblentz was the most important pioneer of IR spectroscopy [13]. In 1905, he reported the results of a thorough investigation of substances with spectra recorded between 1000 and 16,000 nm. It enabled the association of the properties of atomic groups in molecules with specific absorptions in the mid-range IR (2500–50,000 nm) [14]. The chemical compounds vibrate energetically like springs in the presence of energy in the NIR spectroscopy. However, quantum physics constrains the atoms in molecules and allows only a small number of allowed energy states. As a result, chemical bonds with different strengths, such as O–H, C–H, and N–H, require different amounts of energy to vibrate from one level to the next. A spectrum shows this energy change as a sequence of absorptions at different wavelengths [11,15].

One of the most useful features of the IR spectra is the bands between 8500 and 12,500 nm, which are known as "fingerprints" because of the potential to identify many pure chemicals. This range is particularly typical of the molecule under study. While interest in NIR spectroscopy was limited to quantitative measurements of water, some basic chemical compounds, and a very small number of studies of individual proteins, spectroscopy in the mid-range IR experienced another heyday,

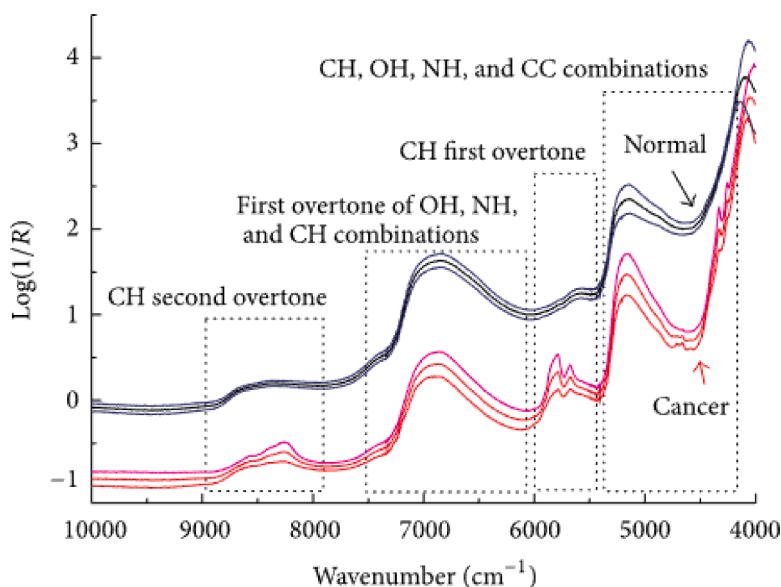


Fig. 1. Mean NIR spectra of healthy control and cancerous samples with an indication of the major NIR-absorbing components (oxy-, deoxy-, and total-hemoglobin, water, and lipids) [12].

especially after World War II. However, no one assumed that it would be useful for characterizing samples, and its application for quantitative analysis was considered too difficult [11,16].

The absorption bands in NIR arise when the energy of the incident light matches the difference between two vibrational energy levels of the molecules. This interaction between the light frequencies and the molecules causes the absorption of some frequencies of the incident radiation, while others may be only partially absorbed or not absorbed at all. The intensity of the absorbed radiation is dependent on the frequency of the incident light source, and this constitutes the absorption spectra of an analyte [17].

The absorption of light by matter occurs when the electrical field of the electromagnetic wave interacts with the electric field of the molecules. For absorption to occur, there must be a change in the dipole moment due to the vibrational movement of the atoms that form the molecular bond or the atoms forming a local group of vibrating atoms [18]. Combination bands, which involve transitions to excited states that involve two vibrational modes simultaneously, are allowed and occur in polyatomic molecules.

The intensity of absorption bands depends on the degree of the dipole change during the vibration of the covalent bonds. Covalent bonds exhibit fundamental vibrational transitions in the region of 3000–4000 nm, while the intensities of the combination absorption bands are about ten times weaker compared to fundamental vibrations and up to a thousand times weaker for the overtone vibrations. The spectral features in the NIR region are dominated by overtones and combination absorption bands [19].

The NIR technique is fast (analysis time varies from seconds to minutes per test), reliable, nondestructive, and inexpensive in terms of cost per test for various applications [20]. In addition, there is no sample preparation or pretreatment, no need for hazardous reagents or solvents, and no disposal problem. These advantages can eliminate sampling errors caused by manual sample handling and reagent contamination. The samples can also be stored for further analysis. It can also be performed by technically untrained personnel on the line or automatically online. In addition, NIR analysis can provide results for many constituents simultaneously by acquiring the NIR spectrum of a sample over a range of wavelengths during a single scan and analyzing the spectrum using multiple calibration equations [21]. The calibration equations are developed through a modeling process using chemometric methods that uses a training set of samples to teach the computer to relate subtle

differences in spectral features to the composition of the sample [22]. A single spectrum can be subjected to many different calibration models to measure any number of constituents.

2. Absorptions in the NIR region

Without the quantum effects, the vibrations of the chemical bonds would be limited to two states, and there would not be enough energy for the vibrations to reach the NIR region. The chemical bonds would only vibrate in the infrared region (700–1000 nm) and the near-infrared region (1000–2500 nm) would not be affected. Therefore, fewer absorptions would be expected in the NIR region if chemical bonds behaved like weak springs. Harmonics and combinations are two phenomena that cause fundamental vibrations to absorb in the NIR region (780–2500 nm) [23]. Harmonic oscillations are a suitable mode to describe harmonics. Consequently, each fundamental vibration leads to a sequence of absorptions at multiple frequencies (the frequency is the reciprocal of the wavelength). Combinations consume more energy and it is more complicated to disentangle because NIR absorptions are more excited than fundamental absorptions [24]. Combinations occur when two or more fundamental absorptions share energy. As a result, many NIR spectra appear uninteresting because of the combined effect of all these absorptions. For example, chloroform (CHCl_3) has only one hydrogen atom. However, this one atom is responsible for all the absorption in its NIR spectrum [25].

The complexity of NIR spectra is often underestimated. Hydrogen is the most common element in NIR spectra, which is a significant generalization. For example, methanol (CH_3OH) has four hydrogen atoms, three of which are equivalent. This spectrum exhibits broad absorption features with small peaks [26]. The analysis of NIR spectra is more complex, but some general conclusions can be drawn. Since the absorption produced by an O–H stretching is different from that produced by a C–H stretching, the overtone series produced by the two types of absorption is also different [27]. Combinations of stretching and bending within a single group are the most common combination bands found [27].

When the complexity of NIR absorption was recognized, most researchers believed that nothing could be learned from NIR spectroscopy analysis. Currently, this point of view is still taught to numerous young scientists. However, with the advances in spectrometers with low-noise spectrometers coupled with chemometric tools, machine learning or

deep learning algorithms, it is possible to create alternatives for functional diagnosis and treatment monitoring using NIR bands [27]. Especially for agricultural samples, NIR analysis can be very useful for quantitative analysis [21]. One advantage of NIR is the ability to use reflected energy instead of solid substances. Hours of analysis can be reduced to minutes, and the same data can provide many analytical conclusions. However, complicated calibrations require many samples and a very good experimental design [21,27]. When a molecule or an atom absorbs light, it makes a transition from one energy level to another. The energy difference between the two levels determines the frequency or wavelength of the absorbed light. In the near-infrared (NIR) region of the electromagnetic spectrum, the energy differences are usually associated with molecular vibrations and rotations. When a molecule or atom absorbs a photon of NIR light, its vibrational or rotational state can change, leading to the emergence of higher energy levels or excited states. These excited states can fall back to the ground state by emitting lower energy photons, often in the form of harmonics and combinations. Harmonics and combinations are vibrational modes that occur at higher energies than the fundamental mode. They result from the simultaneous excitation of multiple vibrational modes. For example, the bending and stretching vibrations of a molecule can combine to form a combination band. These harmonics and combinations can be detected by absorption in the NIR regions of the spectrum where the energy differences correspond to these higher vibrational modes [28]. Access to these harmonics and combinations can provide valuable information about the molecular structure and composition of a substance. NIR spectroscopy is a widely used technique that takes advantage of these absorption features to analyze the chemical and physical properties of a wide range of materials, including foods, pharmaceuticals, and agricultural products. By measuring the absorption of light in the NIR range, this technique can provide information about the chemical composition, moisture content and other properties of the sample [28].

3. Application of chemometrics and machine-learning algorithms

The advances in the use of NIR spectroscopy instruments are mainly related to the development of faster computers and mathematical tools, such as multivariate analysis. The improvement of chemometric analysis, machine learning and deep learning algorithms to characterize chemical components and to discriminate samples has grown with potential applications of NIR spectroscopy potential in the industry (pharmaceutical, forensic, gasoline and diesel fuel, food, and clinical health services). Multivariate analysis tools can untangle the molecular signals of complex samples and provide models for characterizing unsupervised, supervised, and semi-supervised samples. Variations in sample chemistry, the physical nature of samples, and measurement settings are factors that can affect the integrity of samples used to calibrate spectrophotometers for specific NIR applications [16,29]. NIR spectroscopy permits an analytical understanding of these findings, it is imperative an engagement in interpretive spectroscopy [11,23,30]. However, it is important to highlight that algorithms can perform a full spectrum interpretive analysis to release only the final result of interest for each NIR spectroscopy analysis. For this reason, spectral data are often used in NIR spectroscopy analysis routines without any attempt to interpret the chemical information contained in the sample. Chemometrics, machine learning and deep learning algorithms allow the correlation of subtle spectral variations with sample composition, leading to effective detection and quantification of chemical constituents [21, 31].

The development of chemometrics and the blurring of the boundaries between near-infrared and infrared (mid- and far-infrared) are recent advances in measurement methodology in the infrared range (IR). The IR range harbors unique spectral profiles that offer numerous advantages for sample characterization [3,6]. In terms of biological and

biomedical applications, the choice between NIR and MIR spectroscopy will depend on the specific requirements of the application. Besides, it is necessary evaluate the discrimination performed by both biophotonic platforms for each analysis. NIR spectroscopy could be selected to perform non-destructive, real-time analysis of biological samples. Currently, there are portable ATR-FTIR spectrophotometers that are cost-effective. However, there are still no diagnostic analyzes approved by health regulatory agencies using the ATR-FTIR as a diagnostic tool, which demonstrates that there is a difficulty in obtaining adequate accuracy in large-scale studies. [21]. While MIR spectroscopy may be preferred for more detailed structural analysis of specific chemical functional groups in samples at the microvessel level [32]. Ultimately, the relative advantages and disadvantages of each technique will need to be carefully considered in order to choose the most appropriate technique for a given application.

The vibrational modes in the MIR frequencies are often clearer and more distinct than those in the NIR frequencies. While the overtones of the O—H, N—H, C—H, and S—H bands can still be detected in the NIR range, these functional groups are much weaker than the fundamental frequencies in the MIR range. Moreover, the combination of bands, such as the C—O stretch and N—H bend in proteins, creates a crowded NIR spectrum with bands that strongly overlap. This overlapping and complexity in the NIR spectrum is a major drawback as it makes it difficult to interpret and quantify data. However, this characteristic also means that fewer wavelengths are needed for calibration and analysis procedures [33–35].

Although the vibrational bands displayed in near-infrared spectra overlap and can appear vague and poorly defined, chemometric tools and both machine learning and deep learning techniques can be used for data processing and exploration to alleviate these spectroscopic limitations [24]. Furthermore, chemometric analysis and machine learning can reduce the need for time-consuming laboratory procedures. The study of lignin polymers (2270 nm), paraffin and long-chain alkane polymers (2310 nm), glucose-based polymers such as cellulose (2336 nm), amino acid polymers such as proteins (2180 nm), carbohydrates (2100 nm), and moisture has traditionally been performed using conventional NIR spectroscopy (1440 and 1940 nm) [11,23]. The discrimination of these complex samples highlights the potential of NIR spectroscopy even for complex human samples.

The significance of NIR spectroscopy is continuously increasing due to recent advancements that have accelerated this trend. Over the years, it has become evident that NIR spectroscopy can be utilized in innovative ways and provide valuable information that cannot be easily obtained through other techniques [21]. NIR spectroscopy has demonstrated unprecedented success in the industry in a variety of applications using multiple instruments for the analysis of synthetic and natural materials [13], including grains, feeds, baked goods, flour, beverages, feeds, pharmaceuticals, dairy products, hydrocarbons and petrochemicals, fine chemicals, radioactive and hazardous materials, and medical imaging and diagnostics [11,23,30]. NIR spectroscopy has traditionally been used for process control, quality assessment, raw material identification, byproduct identification, and quantitative chemical analysis of complicated mixtures [11,23,30]. The spectrum-structure correlation is used to establish relationship between the instrument response and the reference data (the analyte) and also a more solid scientific basis for multivariate analysis in NIR spectroscopy spectra [11,23,30]. Overall, the extensive applications and knowledge in chemometrics can be successfully applied to NIR spectroscopy, making it a versatile and useful analytical tool for a wide range of applications.

3.1. Spectral preprocessing techniques

The development of analytically referenced calibration models is essential for obtaining accurate multivariate calibrations in NIR spectroscopy. These models provide a link between X, which represents the

instrument response data or spectrum, and Y, the corresponding reference data. It is critical that the model capture a causal relationship between the spectrum-structure correlation and the reference values to truly enable predictive capabilities. Understanding this cause-and-effect relationship forms the basis for informed scientific decision making [11, 29].

In the context of systemic diseases and their associated molecular changes, machine learning algorithms play a critical role in discriminating samples for screening and diagnostic applications using NIR spectroscopy. Effective data analysis involves preprocessing techniques such as filtering or normalization, followed by the application of state-of-the-art algorithms such as deep learning, machine learning models, or chemometric tests. Using these techniques on multicomponent data improves accuracy, resulting in significant increases in sensitivity and specificity, which are essential for accurate disease diagnosis and monitoring.

For chemical analysis using spectral data, statistical and advanced mathematical methods, especially multivariate regression analysis, are commonly used. Univariate spectral analysis has its limitations, so various multivariate regression techniques have been developed to extract meaningful information from spectral data (Fig. 2). These techniques allow simultaneous analysis of multiple wavenumbers or wavelengths, facilitating comprehensive biochemical analysis. The advantage of this approach is that it allows the evaluation of different sample components without the need for spectral resolution or deconvolution [36,37]. Analytically referenced calibration models provide the basis for accurate multivariate calibrations in NIR spectroscopy. Machine learning algorithms, along with preprocessing techniques and advanced mathematical methods such as multivariate regression analysis, help improve accuracy and sensitivity in early disease detection and diagnostic applications. Using these tools, meaningful insights can be gained from spectral data to support scientific decision making and advance our understanding of complex molecular changes in systemic diseases. To develop effective algorithms for discriminating samples, raw spectral data can be preprocessed by one or more specific spectrochemical data analyses, such as (1) truncation in a region of interest, which involves eliminating spectral data outside of a specific range of wavenumbers that are relevant to the analysis. This technique can help remove irrelevant spectral features and improve the signal-to-noise ratio of the data. (2) smoothing filters to remove potential artifacts. Different types of smoothing filters can be used, such as moving average or Savitzky-Golay filters, depending on the nature of the spectral data and the analysis requirements, which can be applied to reduce the effect of noise and other artifacts in the data. (3) normalization is another important technique that can be used to correct for variations in the intensity of the spectra caused by differences in sample preparation or instrumental

settings. Different types of normalization methods can be used, such as scaling by a specific vibrational mode, by total area under the curve or normalization by vector. (4) application of corrections to the spectra, such as rubber band or automatic weighted baseline correction, which are effective at correcting for baseline offsets, or second derivative and normalization-based methods, which can be used to correct for spectral distortions. (5) dimensional reduction of the data is another preprocessing technique that can be used to simplify the spectral data by reducing the number of variables while preserving the important spectral features. Techniques such as principal component analysis (PCA), partial least squares regression (PLSR), or independent component analysis (ICA) can be used for this purpose, and (6) feature selection can be used to identify and extract the most relevant spectral features for discrimination purposes [38–40]. This technique can help reduce the computational complexity of the analysis and improve the accuracy of the discrimination models. Overall, the combination of these preprocessing techniques with analysis algorithms can improve the accuracy and reliability of spectroscopic data. Therefore, spectral data are preprocessed to remove or reduce the contribution of signals unrelated to the analyte, target properties, or sample discrimination (which depends on chemical composition). Preprocessing of raw data reduces chemically irrelevant variations to improve the accuracy and precision of qualitative and quantitative analyses. The main role of preprocessing is to convert the spectrum to the best-fitting condition and ensure that optimal performance can be achieved in later steps. This process is essential to correct physical disturbances, such as light scattering due to different particle sizes, sample thicknesses, optical paths, and random instrumental interference. However, preprocessing techniques also run the risk of introducing undesirable correlations in the noise structure that would negatively affect the quality of the multivariate model. Therefore, preprocessing techniques should be used judiciously and not overused [39].

3.2. Unsupervised and supervised algorithms for spectral analysis

After the preprocessing step, the data are often split into a training set to create the classification/prediction algorithms and a test set to evaluate performance on external samples. Otherwise, the bootstrap method with replacement is preferable because the entire data set is used. NIR spectral analysis involves the use of machine learning algorithms, which can be categorized into two main types: unsupervised and supervised techniques. Unsupervised techniques are used to explore the structure of the data and identify any patterns or clusters within the data without any previous knowledge of the sample composition or properties [41]. Principal Component Analysis (PCA), Cluster Analysis, and Self-Organizing Maps (SOM) are common unsupervised techniques used

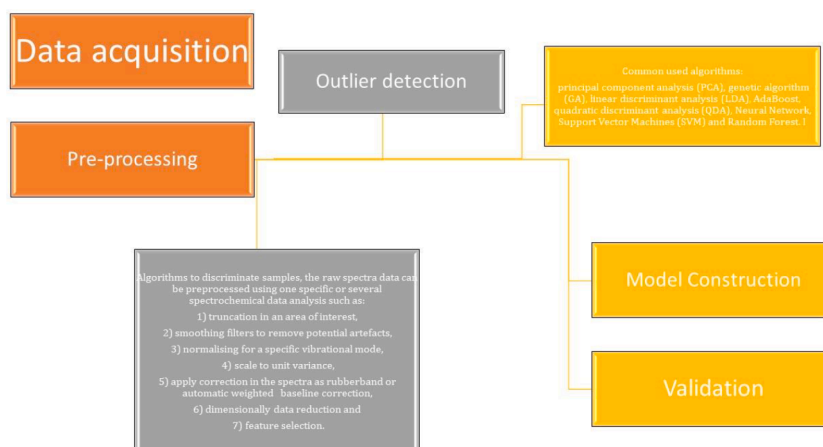


Fig. 2. Spectral data analysis flowchart.

in NIR spectral analysis. PCA is a commonly used unsupervised technique in NIR spectral analysis that can help to reduce the dimensionality of the data while retaining the most important spectral features. PCA works by identifying the principal components, which are linear combinations of the original spectral variables that capture the maximum amount of variance in the data. The Cluster analysis is another unsupervised technique that can be used to group similar samples together based on their spectral characteristics. This technique indicates by assigning each sample to a cluster based on its spectral similarity to other samples within the dataset. Cluster analysis can contribute to identify any natural groupings or trends within the overall data, which can be useful for identifying potential outliers or for selecting representative samples for subsequent analysis. Finally, SOM is an artificial neural network that can be used to visualize and explore high-dimensional NIR spectral data. This algorithm organizes the data into a two-dimensional grid of nodes, with each node representing a specific spectral pattern or cluster within the data. By mapping the high-dimensional data onto this lower-dimensional grid, the data can be visualized and analyzed in a more intuitive way [41–44]. An example of unsupervised machine learning techniques to classify and group a dataset of descriptive variables from 10,442 lung cancer patient records in the Surveillance, Epidemiology, and End Results program database (SEER). The goal is to automatically group lung cancer patients based on clinically measurable disease-specific variables to estimate survival rates. Variables selected for machine learning include number of primary tumors, age, grade, tumor size, stage, and TNM, which are either numeric or can be easily converted to numeric values. The results show that self-ordering maps show the best performance among the analyzed techniques, while k-Means outperforms other simpler classification methods [45]. Supervised techniques, on the other hand, are used when the objective is to develop models for predicting the properties or composition of new samples based on their NIR spectral characteristics [41]. These techniques require a training set of samples with known properties to develop a model for predicting the properties of new, unknown samples. The Linear Discriminant Analysis (LDA), Partial Least Squares Regression (PLSR), Artificial Neural Networks (ANN), and Support Vector Machines (SVM) are common supervised techniques used in NIR spectral analysis. LDA projects the high-dimensional NIR spectral data onto a lower-dimensional space, while maximizing the separation between the different sample groups. The goal of LDA is to identify the linear combinations of the spectral variables that best discriminate between the sample groups, and then use these combinations to classify new, unknown samples. PLSR identifies the linear combinations of latent variables (LVs) that can explain both the variation in the predictor variables (spectral data) and the variation in the response variables (properties of interest), then uses these combinations to develop a model for predicting the properties of new samples. This technique is commonly used in NIR spectral analysis because it is capable to handle highly collinear and noisy spectral data. Differently, ANN and SVM are supervised techniques that can develop nonlinear models for predicting the properties of new samples based on their NIR spectra. The ANN models consist of interconnected nodes, which are organized in layers. The input layer receives the NIR spectral data, and the output layer provides the predicted properties or composition of the sample. In between the input and output layers, there can be one or more hidden layers, which help to capture the nonlinear relationships between the spectral variables and the properties of interest. Lastly, SVM algorithm is a type of machine learning algorithm that can be used to develop models for predicting the properties of new samples based on their NIR spectra, by finding the optimal hyperplane that separates the samples with different properties in the spectral space [41,43,46,47]. Several other machine learning algorithms can also be used in NIR spectral analysis, such as Quadratic Discriminant Analysis (QDA), Partial Least Squares Discriminant Analysis (PLS-DA), Adaptive Boosting (AdaBoost) and Random Forest (RF). These various techniques can be used to explore linear and non-linear relationships, and capture subtle

differences that are nonetheless important for sample characterization. In some cases, these algorithms can work together to improve performance [48–50], in a framework often referred to as ensemble methods. Nowadays, multivariate statistical approaches are used in the study of quantitative structure-activity relationship (QSAR). Linear and nonlinear approaches are used to develop QSAR to solve the difficulties in regression and classification of data structure. Exploratory data analysis methods used for categorization or pattern identification primarily include PCA and hierarchical cluster analysis (HCA). In addition, artificial neural networks (ANN), principal component regression (PCR), partial least squares (PLS-R) and PLS-DA (for discrimination purposes), CART, classification and regression trees are some of the commonly used analysis techniques to extract relevant features to build predictive and classification models. In addition, k-nearest neighbor (kNN), soft independent modeling of class analogy (SIMCA), and support vector machines (SVM) were presented as algorithms for pattern recognition. Moreover, these approaches were used in different contexts, which provided a better understanding of their capabilities [44,51]. One notable application is the differentiation of healthy and tumor tissue during brain tumor surgery, where the combination of hyperspectral imaging techniques (HSI) and ML has proven to be a reliable tool. Therefore, thirteen *in vivo* hyperspectral images of twelve patients with high-grade gliomas (grade III and IV) were selected to train SVM, RF and CNN classifiers. Five different classes were defined in the experiments: healthy tissue, tumor, venous blood vessel, arterial blood vessel, and dura mater. The results show that the overall accuracy (OACC) of the classifiers ranges from 60 to 95%, depending on the specific training conditions. This demonstrates the potential of HSI in conjunction with the algorithms of ML to achieve high accuracy in discriminating tissue types during brain tumor surgery. In particular, the contribution of each band to the overall accuracy is 3.81 times higher than reported in the literature [52].

Overall, the development of machine learning algorithms has opened up new opportunities for analysis and interpretation of NIR spectroscopy data. Proper preprocessing techniques are critical for the success of machine learning algorithms. The choice of the appropriate machine learning algorithm depends on the specific problem and the available data. Therefore, chemometric analysis and machine learning are widely used to overcome these challenges by extracting relevant information from the spectral data and correlating it with the sample properties [53]. As machine learning algorithms continue to improve and evolve, their applications in NIR spectroscopy and other analytical techniques are likely to become even more widespread and impactful.

4. Applications of NIR spectroscopy to cancer and other diseases

A well-ordered vascular network allows adequate nutrient supply for the metabolic needs of cells from healthy tissue. In this condition, the endothelial layer of blood vessels is finely structured with a basement membrane and pericytes [54]. In contrast, in tumor tissue, the high release of proangiogenic factors promotes a chaotic and disorganized network of blood vessels. The blood vessels formed during the excessive release of proangiogenic signaling are less covered by pericytes and present fewer endothelial cell connections, resulting in significant changes in interstitial fluid pressure in tumor tissue. This vascular leakage also interrupts nutrient supply, which can promote glycolytic and hypoxic environments leading to secondary hyperacidity of tumor tissue [54]. High metabolic demand and/or poor blood flow, high hemoglobin and low oxygen saturation are common signs of tumors [6]. These changes in the vasculature of tumor tissues are related to important NIR-absorbing components such as oxy-, deoxy-, total hemoglobin and water (Fig. 3).

4.1. Cancer

These functional changes are related to both cancer diagnosis and

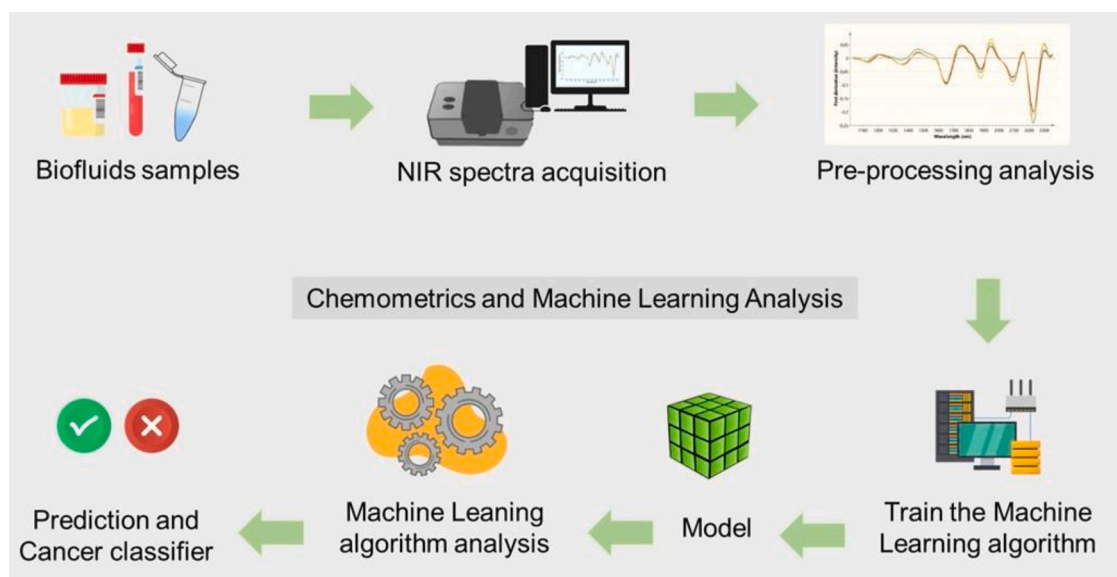


Fig. 3. Pipeline NIR spectroscopy for diagnostic tool for early cancer detection.

therapeutic monitoring. An early cancer diagnosis can greatly improve treatment outcomes. For example, if breast tumors are detected *in situ* or early-stage invasive confined within the breast area, the 5-year survival rate is 90% or higher, and it reduces after spreading to lymph nodes and beyond the breast region. Ionizing or non-ionizing radiological methods, including X-ray mammography, are currently used for cancer diagnosis [3]. Some of these imaging modalities have made significant progress in recent years. However, they are low-throughput technologies that are quite expensive and may expose users to ionizing radiation (such as CT and PET), limiting their widespread use [3,4]. Recently, NIR spectroscopy has gained popularity for cancer functional diagnosis. The light can penetrate deep or large amounts of tissue because there is a known spectral window in the near-infrared range (650 to 950 nm), where tissue absorption is comparatively low (up to several centimeters) [4]. NIR spectroscopy provides a rapid and convenient complement to expensive imaging technologies (such as fMRI, PET, and CT) for measuring optical tissue features at the microvascular level [7]. The ability of NIR spectroscopy to assess tumor blood flow, oxygenation, and oxygen metabolism has already been investigated for use in cancer diagnosis [5]. The identification of differences between tumor and normal blood flow in breast cancer is one example.

NIR spectroscopy can distinguish different tissue types and detect the presence or absence of molecules of interest (e.g., proteins) in a sample. In addition, it can be used to monitor the activity of enzymes in living cells *in vitro* and to measure the levels of various metabolites, hormones, and other small molecules in biological samples *in vivo*. NIR spectroscopy is a powerful diagnostic tool for early cancer detection. It can be used to identify the chemicals that are released when cancer cells grow and spread [55]. NIR spectroscopy can also determine the stage of cancer and whether cancer has spread to other parts of the body [56]. For example, it can determine if cancer cells are present in a tumor in the pancreas. NIR spectroscopy can be used in conjunction with imaging and other tests to make a more detailed and accurate cancer diagnosis. It can also help clinicians select the most likely effective treatment for each patient [3,4]. In this way, clinicians can track disease progression and identify treatments that are most effective at different stages of the disease [57]. For example, detecting breast tumors when they are still localized results in a 5-year survival rate of 90% or more. An early cancer diagnosis can significantly improve treatment outcomes. Due to their high metabolic demand and/or poor perfusion, high hemoglobin levels and low oxygen saturation are often signs of tumors [6].

NIR spectroscopy also acquired attention for cancer treatment

monitoring. The lack of appropriate technology prevents routine measurement of tumor hemodynamics and metabolism during cancer treatment [6]. In addition, various cancer treatments, including chemoradiotherapy in human malignancies and antivascular therapy and photodynamic therapy in mouse tumors, have been studied with NIR spectroscopy instruments in animals and humans, suggesting the ability to reflect the metabolic tissue rate of oxygen [4]. Several pilot human studies have demonstrated non-invasive clinical assessment of hemodynamic responses in various malignant tumors (e.g., breast, head and neck, prostate) during different treatments such as chemotherapy, radiotherapy, and photodynamic therapy. The results of these clinical studies show that tumors exhibit significant hemodynamic changes during the administration of therapy and in the first days or weeks after treatment [58,59]. Prediction and timely optimization of cancer therapy could benefit from such findings. Optical technologies such as NIR spectroscopy are noninvasive, rapid, portable, affordable, and continuous. They offer several significant advantages over one-time fMRI, CT, or PET testing. Optical technologies can easily detect these early and constant changes. In addition, these optical tests do not pose significant risks to patient populations [58,59].

The NIR spectroscopy platforms have been associated with other cutting-edge technologies for diagnostics and to monitor the disease and its treatment. In this framework, the bimodal probes with NIR-I fluorescent dye were evaluated focusing on related nanoprobe [60]. These probes allowed us to achieve synergistic imaging while overcoming the limitations of each imaging modality [61]. These NIR-I fluorescence-based bimodal probes are used for cancer theragnostic [60]. NIR (650–1700 nm) bioimaging has become an effective tool for tumor diagnosis. Due to its high quantum yield and favorable biocompatibility, NIR-I fluorescence imaging (650–950 nm) has attracted particular interest. Fluorescent probes for NIR-I imaging have recently undergone many modifications, as their relatively low penetration depth limits some of their biological applications [60].

The application of machine learning models in clinical bioimaging discrimination has become very important due to their potential to improve public and private health services [62,63]. The lack of appropriate technology prevents routine measurement of tumor hemodynamics and metabolism during cancer treatment [6]. Ionizing or non-ionizing radiological methods, including X-ray mammography, are currently used for cancer diagnosis [3]. Some of these imaging modalities have made significant progress in recent years. However, they are costly technologies that may expose users to ionizing radiation

(e.g., CT and PET), so their uptake is low [3,4]. On the other hand, NIR spectroscopy has recently gained popularity for functional diagnosis and treatment monitoring of cancer.

4.2. Changes in physiological profile coupled with other cutting-edge techniques

In another study, the phytochemical composition and variability of acorn morphotypes of *Quercus ilex* were investigated using NIR spectroscopy and mass spectroscopic techniques. According to NIR spectroscopy literature, starch was the most abundant substance (50–60%). Untargeted metabolomics analysis was performed using UHPLC-QToF analysis, and 192 metabolites were annotated. This early research provides new opportunities for the search for bioactive compounds, molecular markers of variability and stress response, and secondary metabolites. In addition, some of the described compounds can be tested for their antibacterial, insecticidal, allelochemical, antioxidant, tumor inhibitory, proangiogenic and other interesting properties [64].

Recently, the potential combination of NIR and mid-infrared (MIR) spectroscopy for the detection of cancer and other disease markers in serum samples from patients has been investigated [33]. Serum is an important candidate in proteomics research because it may contain important biomarkers for disease development and health status. However, due to the wide dynamic range of concentrations of individual protein and peptide species in serum, it is difficult to analyze several significant diagnostic indicators that have been identified in the circulatory proteome and low molecular weight (LMW) peptidome [65]. Serum proteome profiling is a well-known way to find important biomarkers associated with many types of cancer, diabetes, atherosclerosis, and neurological diseases. For example, the detection limit of glycine in serum is improved by the multimodal technique, including MIR spectroscopy with ATR (attenuated total reflectance) and NIR spectroscopy technologies [65]. This proof-of-concept work shows that the combination of NIR and MIR spectroscopies can facilitate the detection of important chemicals in components with low molecular weight fractions (LMWF) [65].

Both NIR and MIR spectroscopies are fast and cost-effective platforms, requiring a simple sample preparation potential and also accurate quantitative analysis using univariate, machine learning, and deep learning analysis for biomedical applications. However, NIR presents suitability to interrogate with moist samples and MIR should be used with dry samples due to the great impact of water in the MIR profile. Besides, NIR spectroscopy is capable to determine functional group properties of tissues at the microvessel level and MIR spectroscopy acts on the surface of the sample [21].

Recent developments in noninvasive continuous radiofrequency measurements ABP can be used with NIR spectroscopy or rTCD to make continuous and noninvasive measurements of cerebrovascular reactivity. These noninvasive measurements can be performed during treatment in patients after a mild traumatic brain injury (TBI), including during out-of-hospital follow-up [66], bypassing traditional intermittent procedures and expensive neuroimaging-based measurements available only in specialized centers. Nonetheless, this highlights the potential of NIR to detect vascular changes in tumor tissues.

The effectiveness of early warning systems, outbreak investigations, and microbiological surveillance of consumer products depends on the rapid identification and strain characterization of pathogens that pose a threat to consumer health and safety. Dieckmann et al. [67] found ten isolates in two bottles of the product. They were identified as *Klebsiella oxytoca* and subtyped using near-infrared Fourier transform (FT-NIR), Raman spectroscopy, Fourier transform infrared (FT-IR), and matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI TOF MS) [67]. A high degree of agreement was found between the classification results of these phenotype-based techniques and those of the DNA-based techniques, pulsed-field gel electrophoresis (PFGE), multi-locus sequence typing (MLST), and single nucleotide

polymorphism (SNP) analysis from whole-genome sequencing (WGS) data. Therefore, to ensure that consumer products are safe and their origin can be traced, it may be helpful to have a range of analytical methods for rapid, reliable, and cost-effective microbial typing [67].

4.3. Renal and endocrine diseases

The current gold standard for collecting important diagnostic and prognostic data in individuals with diabetic nephropathy is the histological evaluation of renal biopsies. On the other hand, the clinical need for novel biomarkers that allow early diagnosis and risk classification remains unmet. Bruyne et al. [37] investigated the potential of using NIR spectroscopy to identify a biochemical signature associated with diabetic nephropathy, as biochemical changes in tissues must occur before the clinical or morphological presentation of the disease [68]. NIR spectroscopy was used to analyze kidney tissue sections, following soft independent modeling of class analogies (SIMCA) and principal component analysis (PCA). The biochemical signature of DN permits discrimination between tissue sections from people with normal histological results ($n = 27$) and those from people with DN ($n = 26$) [68]. Some spectral changes induced by carbonylation and glycation processes appeared to be similar to those found in DN patients. In addition, the spectral pattern was sometimes markedly restored after treatment with the deglycating enzyme fructosamine 3-kinase. Laboratory measurements of diabetic and uremic burden, such as hemoglobin A1c, urea, creatinine, estimated glomerular filtration rate, and proteinuria, correlated significantly with NIR spectral characteristics. The proposed method could be useful for histopathological examination, especially in monitoring renal biopsies after transplantation, and also to prevent or delay the development of renal disease [68] (Fig. 4).

As we have already seen, NIR spectroscopy can also be used in biospectroscopy. It has been used as a diagnostic tool for early diagnosis of human diseases. Biomedical research has found that spectrochemical techniques combined with computational analysis contribute to diagnosis, classification, prediction of outcomes, stratification of treatment, and modulation or monitoring of disease and its treatment. Fourier transform near-infrared spectroscopy (FT-NIR) is one of the main focuses of standardization methods that can be used for a wide range of spectral techniques [38]. NIR spectroscopy has recently been used to detect and monitor changes in glucose levels in the blood and urine of diabetic patients [70]. The capability of NIR spectroscopy to detect these chemical markers can help improve the management of diabetes by providing timely treatment to patients when their blood glucose levels become too high or too low [70].

4.4. Neurological diseases

Multivariate classification techniques using NIR spectroscopy have been used to monitor the progression of human diseases [71]. The spectral resolution of NIR spectroscopy makes it an excellent tool for early diagnosis and monitoring effectiveness of therapeutic interventions. It was developed a protocol for multivariate classification analysis of vibrational spectroscopy data [39]. This method was validated on a set of spectroscopic data obtained in a clinical trial (NCT04867616) involving subjects with mild to severe Alzheimer's disease (AD) or mild cognitive impairment (MCI). Using this method, it was possible to detect individuals with MCI in the early stages of AD before clinical symptoms appeared. It was the first study to demonstrate the feasibility of using NIR spectroscopy to diagnose early-stage AD [72, 73].

5. Challenges and future prospect

Routine measurement of the functional properties of disease biomarkers requires further development in R&D and education despite the enormous potential and use of NIR spectroscopy for early diagnosis of

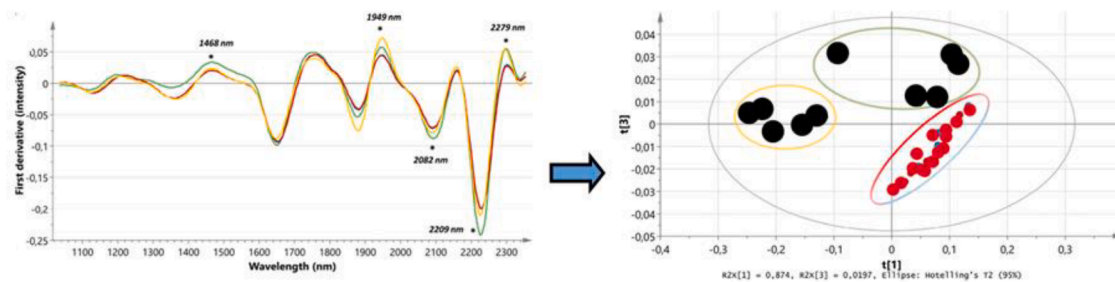


Fig. 4. The image corresponds to a kidney biopsy sample being analyzed using NIR spectroscopy to identify a biochemical signature associated with diabetic nephropathy. This non-invasive technique has the potential to provide early diagnosis and risk classification for the disease, as well as to monitor renal biopsies after transplantation and prevent or delay the development of renal disease. The study conducted by Bruyne et al. [69] found that the biochemical signature of DN permits discrimination between tissue sections from people with normal histological results and those from people with DN, and that laboratory measurements of diabetic and uremic burden correlate significantly with NIR spectral characteristics.

critical diseases such as cancer, diabetes, atherosclerosis, and neurological disorders. For proper use of NIR spectroscopy in disease diagnosis, it is important to develop calibration models that take into account multiple factors, such as changes in the environment, differences in the preparation of individual samples, and the best way to preprocess the spectral data. However, this essential requirement is often overlooked or underestimated. It is of utmost importance to the medical community and patients that urgent and emergent conditions be diagnosed as quickly as possible. Current patient volumes, disease complexity, and complex patterns in biological samples contribute to the challenges of using NIR spectroscopy in healthcare.

Compared to other methods, bioanalytical research and medical diagnosis with NIR spectroscopy are still in late stages for clinical application. Infrared and Raman spectroscopy remain strong competitors in these areas. However, as this review shows, NIR spectroscopy is gradually expanding into new areas of medical diagnosis. IR and Raman spectroscopy are often considered techniques with higher chemical specificity than NIR spectroscopy, but recent advances have made NIR spectra easier to understand. This is especially important when working with biological samples that are chemically complex. As data analysis methods improve, the use of NIR spectroscopy in bioapplications is expected to increase in the coming decades. In the future, NIR spectroscopy and other sensor technologies, data analysis methods, and online approaches could be used to further improve the ability to detect diseases early.

6. Conclusion

In this review, we attempted to evaluate the application of NIR spectroscopy in the identification of biomarkers for several diseases, particularly cancer. The measurement of blood flow in tumors using NIR spectroscopy is an outstanding platform for early cancer detection. Red blood cell movements are susceptible to speckle variations of diffuse NIR light in tissue, which can be measured. Tumor blood flow has been measured continuously in several pilot studies before, during, and after cancer treatment. These results have been shown to provide essential factors for distinguishing between benign and malignant tumors and for predicting the course of therapy. NIR spectroscopy, in conjunction with machine learning algorithms and chemometrics, is expected to become an important method of choice for managing cancer therapy as more clinical applications are performed on large patient populations. It will undoubtedly provide important information about cancer physiology that will aid prognosis.

CRediT authorship contribution statement

Rui Vitorino: Supervision, Funding acquisition, Conceptualization, Writing – review & editing. **António S. Barros:** Writing – review & editing. **Sofia Guedes:** Investigation, Conceptualization, Writing –

original draft. **Douglas C. Caixeta:** Writing – review & editing. **Robinson Sabino-Silva:** Supervision, Methodology, Conceptualization, Writing – review & editing.

Declaration of Competing Interest

The authors declare no conflict of interest.

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